

Part 1: Group Formation and Initial Presentation

Objective

The objective of this part is to form a group, select a dataset, choose a topic or define your goal, select appropriate visualization techniques, and justify the algorithms you plan to use. **You are not required to implement anything yet at this stage**, just plan for your project. This will set the foundation for your data science project. You will also need to prepare a 10-minute presentation.

Instructions

1. **Group Formation:** Form groups of 2-3 students. Each group should select a team leader who will coordinate tasks and ensure deadlines are met. Please enroll yourself into groups on eClass (You can find the activity under the Week 5 module – Practice: “Project Part 1: Form a Group”).
2. **Define Topic/Goal:** Clearly state the hypothesis or goal of your project. For example, "Analyze the impact of traffic volume on accident rates in Toronto." Ensure your hypothesis is specific, measurable, and researchable. A well-defined hypothesis will guide your analysis and help you stay focused on your question.
3. **Dataset Selection:** Choose a dataset from the City of Toronto Open Data Portal or any other reliable source. Examples of datasets include Toronto Traffic Volume Data, Toronto Public Health Data, and Toronto Crime Rates. Ensure the dataset is relevant to your topic or your goal and contains sufficient data for analysis.
4. **Dataset Preprocessing:** Describe the preprocessing steps you plan to take to clean and prepare your data for analysis. This might include handling missing values, discretize the data, or convert text values to numerical values.
5. **Algorithm Selection:** Choose appropriate algorithms, such as linear regression, logistic regression, or clustering. Justify why these algorithms are suitable for your project. For example, "We will use linear regression to predict accident rates based on traffic volume and clustering to identify accident hotspots." Selecting the right algorithms is essential for accurate analysis and meaningful results.

6. **Visualize the Data:** Decide how you will visualize the data, such as using bar charts, scatter plots, or heatmaps. Justify your choice of visualization techniques. For example, "We will use scatter plots to show the relationship between traffic volume and accident rates, and heatmaps to visualize accident hotspots." Effective visualizations will help you communicate your findings clearly and effectively.

Instructions for Presentation for Part 1

Prepare a 10-minute presentation covering the above points. Include a brief introduction to your dataset and its relevance. Use slides to organize your presentation and include visual aids. Practice your presentation to ensure it fits within the time limit. All group members must participate in the presentation and each group member will take care 2-3 points below. Please create a clear outline for the presentation and list your names next to the topics you will be presenting.

Presentation Structure:

1. **Introduction:** Introduce your group members. Provide an overview of your chosen dataset, including its source and key attributes. For example, "Our group consists of [names]. We chose the Toronto Traffic Volume Data from the City of Toronto Open Data Portal, which includes traffic counts from various locations across the city."
2. **Hypothesis/Goal:** Clearly state your hypothesis or goal and explain why this is an important question to investigate. For example, "Our hypothesis is that higher traffic volumes lead to increased accident rates. Understanding this relationship can help in designing better traffic management policies."
3. **Dataset and Preprocessing:** Describe the dataset and the preprocessing steps you plan to take. For example, "We will clean the data by removing missing data to ensure the accuracy of our analysis."
4. **Ethics Challenges:** Is there a bias issue in your data? Clarify who owns the data and how it can be used.
5. **Algorithm Selection:** Explain the algorithms you plan to use and justify why these algorithms are appropriate for your data and hypothesis. For example, "We will use linear regression to quantify the relationship between traffic volume and accident rates, as it allows us to predict accident rates based on traffic volume. Additionally, we will use K-means clustering to identify accident hotspots, which will help us understand the spatial distribution of accidents."

6. **Visualization:** Describe the visualization techniques you plan to use and justify your choices with examples. For example, "We will use scatter plots to identify trends between traffic volume and accident rates, and heatmaps to highlight areas with high accident rates. Scatter plots will help us visualize the correlation, while heatmaps will provide a spatial representation of accident hotspots."
7. **Conclusion:** Summarize your plan and next steps. For example, "In summary, we will analyze the relationship between traffic volume and accident rates using linear regression and identify accident hotspots using clustering. Our next steps include data preprocessing, algorithm implementation, and visualization." Open the floor for questions and feedback. For example, "We welcome any questions or feedback on our proposed approach."

Tips for a Successful Presentation:

- Ensure each group member clearly understands their part of the presentation.
- Make eye contact with your audience and speak confidently.
- Use slides effectively to support your points, but avoid overcrowding them with text.
- Do NOT read from the annotations.
- Practice to ensure your presentation fits within the 10-minute limit.

Teamwork

Teamwork is a critical skill in computer science, as most real-world software projects involve collaboration. Working in teams allows you to tackle complex problems more effectively, combine diverse ideas, and learn from each other's strengths. It fosters better problem-solving, encourages innovation, and helps you practice communication skills—key attributes in any tech career. Additionally, teamwork helps manage workloads efficiently and builds a support network for learning.

Suggestions for Effective Teamwork:

- **Set clear goals:** Define the tasks and divide responsibilities based on individual strengths.
- **Communicate regularly:** Consistent communication is key. You are highly encouraged to complete the work during the lab time, as you will receive support from the TA. If you are unable to complete it, you can use any preferred platform

(e.g., Discord, Slack), but make sure to check in regularly to stay on track and share updates.

- **Be open to feedback:** Constructive feedback is essential for improving your work.
 - When giving suggestions, do so professionally and supportively. Expressing frustration or anger toward teammates is not acceptable and will not be tolerated in this course.
 - When receiving feedback, don't take it personally. Remember that input from others is meant to strengthen your team's overall performance.
- **Manage conflicts positively:** If disagreements arise, focus on finding solutions rather than dwelling on the problem.
- **Stay accountable:** Follow through on your assigned tasks and assist your teammates when needed.
- If you're unable to resolve an issue within the team, please talk to your TA. If the TA cannot resolve it, please report them to me with the designated contact method posted on eClass.

Project Part 1 Submissions

You should submit two files:

- Your presentation in the format of “.pdf”.
 - Your presentation will be graded for CLO3-A and CLO3-B.
 - The 1 point for CLO3-B is graded individually. You will receive the point if you participated in the presentation. If you have accessibility needs, please contact the teaching team for alternative arrangements.
 - You can take a look at the grading scheme on eClass for CLO3-A (you can access it where you submit your work on eClass).
- Submit a text file (.txt) indicating whether each group member completed their assigned work. Free-riding is discouraged. If any member did not actively participate in the project (e.g., avoiding collaboration by working on other courses during lab time or being unresponsive), please report it in the file. We will follow up with the reported member for clarification, and in serious cases, they may receive no points for any part of the project.

Please submit your files to “Project Part 1: Presentation”. You can submit your files as many times as you want.